

Laser Distance Module-LDJG Series

Technical Data Sheet



Introduction

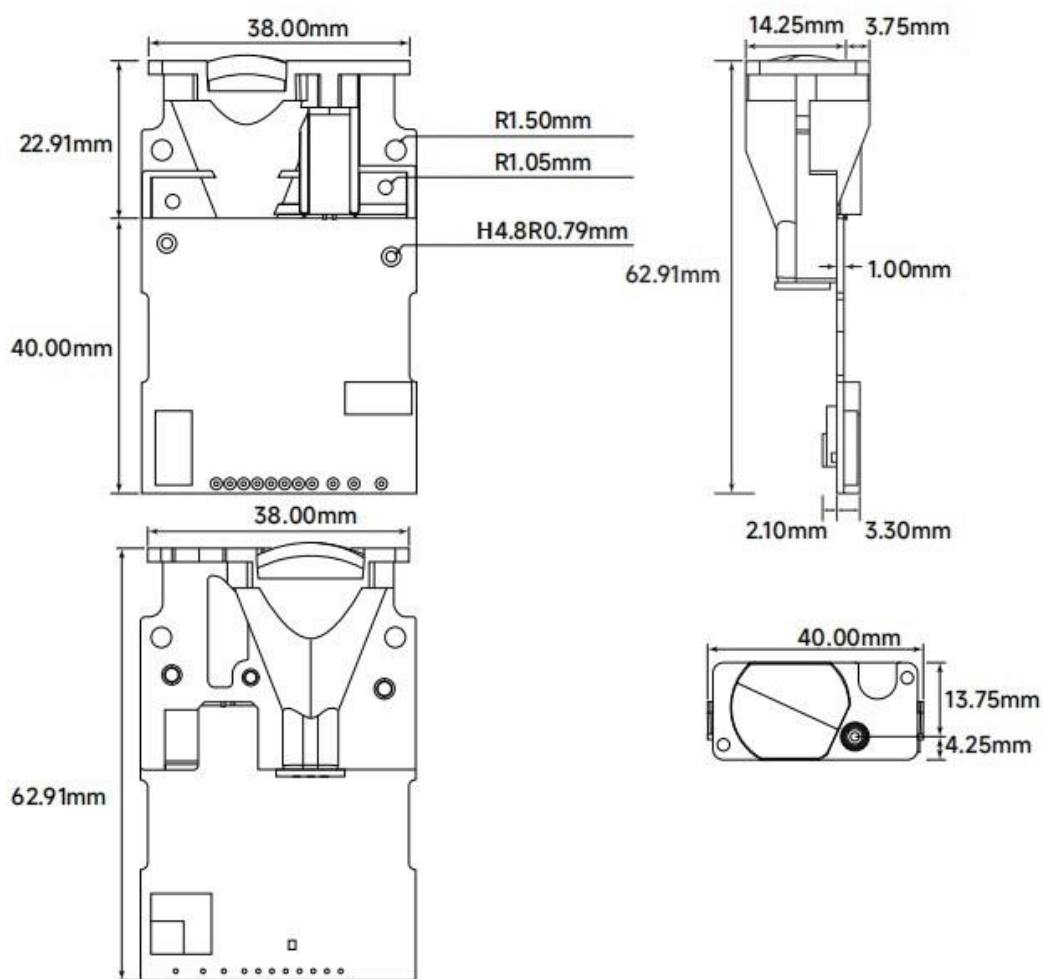
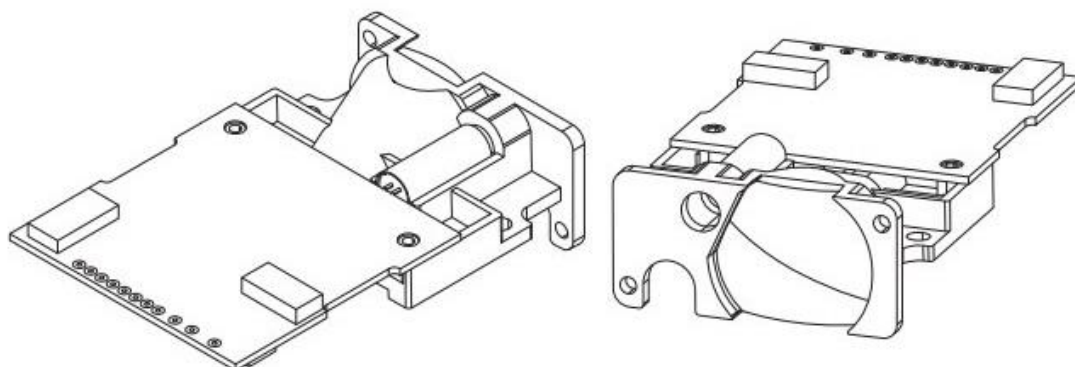
LDJG-100 is a high-precision laser distance module, independently developed by Chengdu Meskernel Integrated Technology Co., Ltd. The extremely small and lightweight features (63*40*18mm, 14g) make it suitable for size and weight limited scenario and convenient for users to integrate into their systems. With low power consumption, stable performance, high accuracy and fast measurement speed, the module has been widely used in industrial automation fields.

Meskernel provides a series of customization service including but not limited to range, accuracy, baud rate, frequency, communication interface, temperature, etc., if you have any customization request, please feel free to contact us.

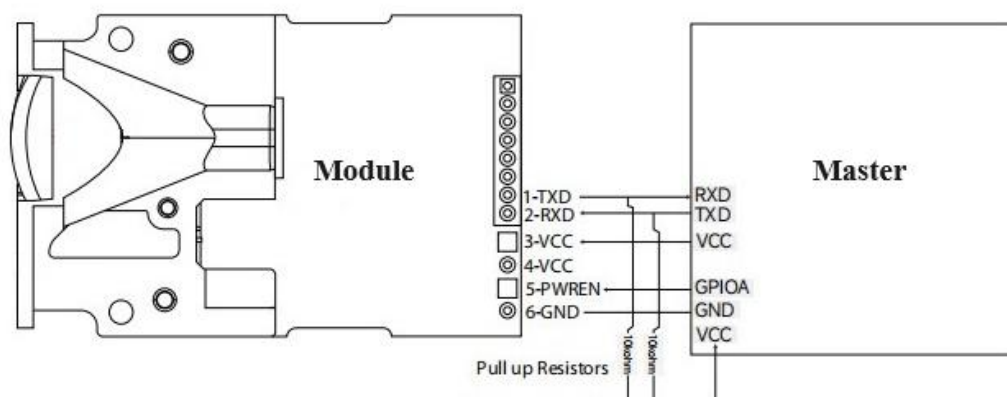
Specification

Optical specifications	
Part No.	LDJG-100
Range	0.03-60m/80m/100m (Reflectivity: 1.0, Interference light intensity: 3kLux)
Accuracy	±5mm (Reflectivity: 0.2-1.0, Interference light intensity: 1kLux)
Frequency	20Hz (Reflectivity: 1.0, Interference light intensity: 1kLux, Distance: 0.03-10m)
Laser Type	520-530nm, Visible Red
Laser Class	Class II (IEC 60825-1:2014), <1mw
Laser Spot	<8mm@10m <20mm@20m <40mm@40m
Measurement Method	Phase-shift principle
Electrical specifications	
Input Voltage	DC 2.5V~3.6V, 3.3V is recommended
Current	Standby: < 60mA@3.3V Laser on: < 220mA@3.3V Continuous measurement mode: <250mA@3.3V
Communication specifications	
Communication Level	TTL (3.3V)
Interface	USART, RS232, RS485
Baud Rate	Auto detection: Default 115200 bps, can be set to 9600~115200 bps Fixed baudrate: Default 115200 bps, can be customized as 4800 bps, 9600 bps, 19200 bps, 38400 bps, 115200 bps
Other specifications	
Dimensions	63*40*18mm (L*W*H)
Weight	14g±1.4g
Working Temperature	0~40 °C (32-104 °F)
Storage Temperature	-25~60 °C (-13~140 °F)
Certification	CE / FCC / RoHS / FDA
Warranty	12 months

Appearance and Dimension



Pin Layout and Definition



PIN No.	PIN Name	Function	Default Value	Description
1	TXD	Output	High	Serial port transmit pin, default open drain
2	RXD	Input	High	Serial port receive pin
3	VCC	Power input	Decided by power supply	Power supply input requirement: >300mA@DC 3.3V
4	VCC	Power input	Decided by power supply	Power supply input requirement: >300mA@DC 3.3V
5	PWREN	Input	Low	Power on enable pin, high level trigger, VIH=2.0V, VIL=0.5V
6	GND	Ground	Decided by ground	Power supply ground

Note

- Under poor conditions (such as: dark color, bad reflectivity or atmospheric visibility, strong ambient light, etc.), the measurement accuracy will have an error of $\pm 1\text{mm} + 50\text{PPM}$.
- Please use the target plate when there is strong sunlight or the target with poor reflection.
- Before using the product, please read the relevant chapters of installation and operation carefully to prevent damage to the sensor, please contact us if you need more information.
- Free of Responsibility: INCORRECT using condition and storage condition, and service life of the module is exhausted during warranty period, the warranty will be TERMINATED.